

REV2 REFERENCE / REDESIGN IN PROGRESS

22 September 2026: the current print order is N1 nose only.

Use Manufacturing/Nose_Print_N1/A59_Nose_N1_mm.stl and Nose_N1_Print_Guide.pdf. Quantity one, millimetres, 100%.

The following pages retain the old Rev2 design for reference. They are not the current full-aircraft manufacturing instructions. Their original page numbers remain unchanged; this notice is an additional page.

Superseded for the new design

Landing gear and steering system; the old 42-piece print order; A59 cover screw holes, columns, spacers and shaped washers; the large nose pitot-access notch. Do not fabricate these from the old instructions.

N1 restores the original upper nose shell. It uses a proposed simple ledge/tape attachment and a probe passage drilled to actual hardware during assembly. It retains the original rear mating envelope.

The rest of the aircraft will be simplified around the plywood skeleton and foamboard. Remaining mounts, control construction, belly protection, installation details and mass/CG estimates still require that redesign. Do not use the old mass, CG or gear clearance numbers for the new configuration.

Existing cutting templates are retained until a reuse/change list is complete. The N1 release is a nose-only handoff, not a completed redesign of every file in this manual.

Electronics build guide

Matek F405-WING V2 / ArduPlane / FlySky / five servos with nose steering

For the **Electronics Team**. Follow the steps in order and record actual readings in the build record.

Remove the propeller for wiring, firmware, calibration, servo and failsafe work. Refit it only for a separate controlled propulsion test after bench checks pass.

Equipment	Build assumption
Flight controller	Matek F405-WING V2; not V1, WTE or TE
Navigation	Matek M9N-5883 GNSS + compass; 32 x 32 x 10 mm, 14.5 g
Controls	Five TowerPro MG90S: two ailerons, two V-tail controls, nose steering. Nominal 13.4 g each.
Power	GenX 4S 5200 mAh: 138 x 43 x 33 mm, 512 g. ReadytoSky 40 A 2-4S with 5 V / 3 A BEC.
Propulsion	Emax ECO II 2807 1300 KV + Gemfan 9045 CL three-blade; selected 4S /40 A setup
Radio / airspeed	FlySky FS-i6X / FS-iA10B; Matek ASPD-4525; SiK 433 MHz telemetry.

Work in this order

Identify parts -> build and meter harnesses -> configure firmware -> connect radio and sensors -> centre and test servos -> test failsafes -> install and balance -> qualify propulsion. Save a parameter file after each successful stage.

Five servos: two ailerons, two tail controls and one nose steering. Establish control throws, flight CG, speeds, gains and return altitude from the completed aircraft and measured tests.

Match the confirmed hardware

Connector shape and wire colour are not a pinout

Component	Connection and check
M9N-5883	5V to FC 5V; G to G. Module TX to RX3, RX to TX3. Compass DA to DA2, CL to CL2. Mount level with arrow forward; use manufacturer ArduPilot orientation and verify heading. [18]
ASPD-4525	5V to FC 5V, SCL to CL2, SDA to DA2, G to G. ARSPD_TYPE=1; ARSPD_BUS=1. GPS compass and airspeed share I2C2 in parallel. [1,6]
ESC signal plug	This ReadytoSky listing specifies a 5 V / 3 A BEC. Depin and insulate its BEC output lead; retain signal and signal ground. Verify the actual plug before powering. Do not join it to the FC servo or 5 V rail.
MG90S x5	Use the FC Vx 5 V setting and initial 50 Hz PWM; do not select 6 or 7.2 V. Centre and limit each servo before attaching linkages. Published stall torque is not a continuous load rating. Load-test the whole rail. [19]
Battery / connectors	Design pack: 138 x 43 x 33 mm, 512 g. Meter connector polarity and record charger limits. Never connect the 4S main lead to a receiver, GPS, servo or 5 V pad.

Make a connector record

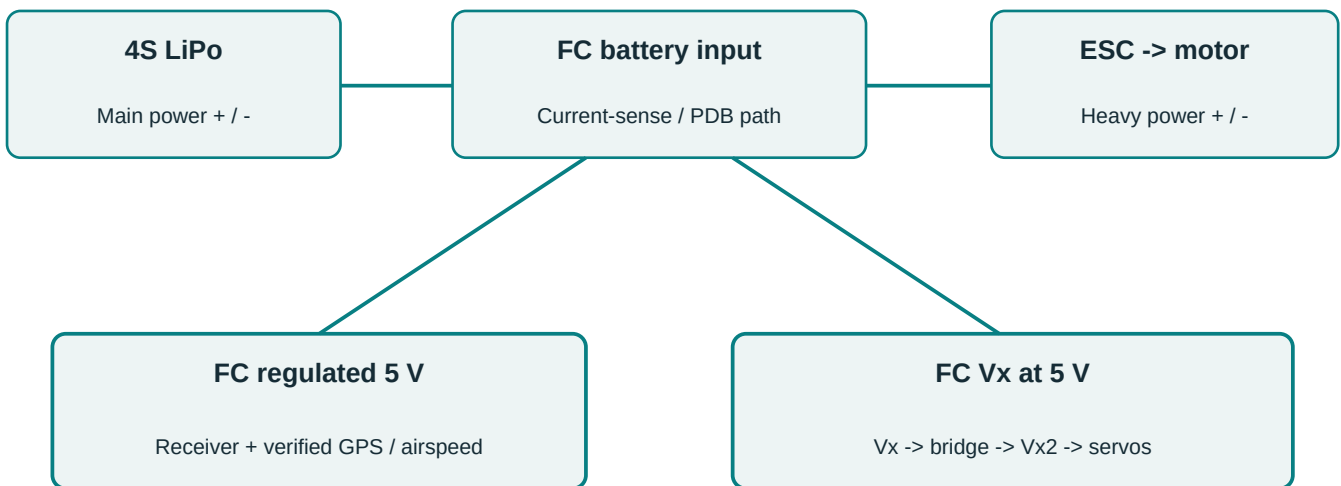
Draw each plug from a stated face. Number its cavities and label signals. With power removed, continuity-check both ends and label the harness.

Pad names below identify functions. Verify connector cavity numbers from the exact hardware manual before soldering.

Power architecture

One positive source per rail; common signal ground

Route battery and ESC power through the FC power-distribution pads. Match the V2 board layout [1].



ESC signal + ground -> FC S1 + G

ESC BEC output (if present) -> insulated; never joined to FC rails

Build and meter

1. Fit strain-relieved battery and ESC power leads to the documented pads. Keep high-current loops short. Select connectors and wire for measured load; insulate exposed joints.
2. Keep electronics 5 V and servo Vx positives separate. S5-S9 use Vx2: bridge Vx to Vx2 for this internal-BEC plan. Check before plugging in servos. [1]
3. If tests require an external BEC: remove the bridge, verify isolation, power Vx2 from that BEC and move all servo positives to that rail. Retain grounds; never join two BEC positives.
4. With peripherals unplugged, check polarity and hard shorts. First power through a suitable current limiter, then meter 5 V and Vx/Vx2 at far connectors. A limiter may brown out under servo load; it does not qualify flight power.

Wire the signal harness

FC outputs and transmitter channels are independent

FC connection	Destination / setting
S1 + G	ESC PWM signal + ground. SERVO1_FUNCTION=70 (Throttle). Isolate BEC output lead if present.
S2	Unused, SERVO2_FUNCTION=0; leave empty.
S3 + Vx + G	Left aileron. SERVO3_FUNCTION=4 (Aileron).
S4 + Vx + G	Right aileron. SERVO4_FUNCTION=4 (Aileron).
S5 + Vx2 + G	Left tail. Initially SERVO5_FUNCTION=79 (V-tail left). Use the V-tail direction checks; physical motion governs.
S6 + Vx2 + G	Right tail. Initially SERVO6_FUNCTION=80 (V-tail right). Use the V-tail direction checks.
S7 + Vx2 + G	Nose steering A39: SERVO7_FUNCTION=26 (GroundSteering). Omit only with a deliberately fixed nosewheel.
S8-S9 / LED	Unused; disconnect. Check SERVO8-10_FUNCTION for unwanted assignments.
RX2 + 5 V + G	FS-iA10B iBUS servo-output signal + supply + ground. Not SENS telemetry port. BRD_ALT_CONFIG=0.
RX3 / TX3 / supply / G	M9N-5883: module TX -> RX3, RX -> TX3; 5V -> 5V; G -> G. SERIAL3_PROTOCOL=5 (GPS).
DA2 / CL2 / supply / G	M9N compass DA + ASPD SDA -> DA2; both CL/SCL -> CL2. Both on regulated 5 V and G. Check both devices detected after connection.

Custom output map: write and read back every assignment before connecting the ESC. S1 is throttle only after configuration. [2-4]

Start with supported 50 Hz PWM. Use conventional PWM for this ESC; leave DShot disabled. Check shared timer groups before changing rates. [2]

Firmware and first connection

Mission Planner on Windows; menu names can change

Install the matching target

1. Confirm **F405-WING V2**. Connect a data USB cable with battery, ESC and servos disconnected.
2. Install stable **ArduPlane / MatekF405-Wing**, version 4.4 or newer. Do not select Copter, INAV, F405-STD or F405-TE. [1]
3. With an ArduPilot bootloader, load the matching .apj. Otherwise follow the board DFU first-install procedure using firmware with its bootloader. [2]
4. Record target and version. Save initial parameters and inspect sensor messages. Insert a supported logging microSD with power off.

Configure before powering outputs

Configure: write page 4 settings in Full Parameter Tree/List. Reboot and read back. Use Mandatory Hardware for sensors, radio and modes; Servo Output for servos.

Do not bulk-import an internet parameter file. Keep arming checks enabled. Resolve each failed check instead of setting `ARMING_CHECK=0`. Save the parameters actually read from this board.

Recovery route

Recovery: disconnect propulsion power. Reconnect USB and restore the last known parameter file. Repeat sensor, output and failsafe checks after firmware changes.

Nose steering: centre A39 with `SERVO7_TRIM`; set safe endpoints and reversal so right rudder steers right. Push-test with motor off. Set `GROUND_STEER_ALT` deliberately. Stabilized steering needs healthy heading and `STEER2SRV_P` based on measured turning diameter. [14]

Radio input and modes

Plain controls from FS-i6X; mixing inside ArduPlane

Create an airplane model; select AFHDS 2A and bind the receiver. Disable transmitter V-tail/elevon mixes and centre trims. Use the receiver iBUS servo port with regulated 5 V. [5]

Input	Intended function
CH1	Roll; RCMAP_ROLL=1
CH2	Pitch; RCMAP_PITCH=2
CH3	Throttle; RCMAP_THROTTLE=3
CH4	Yaw; RCMAP_YAW=4
CH5	Three-position mode switch; FLTMODE_CH=5. Measure its actual detent values.
CH6 optional	Dedicated arm/disarm only after verifying the installed firmware option and behaviour; never share with mode control.

Calibrate

Calibrate inputs: power RX from verified FC 5 V. Check each stick moves the intended bar. Sweep sticks and switches; finish at low throttle with other sticks centred. Save. [8]

Input directions: right roll, pitch up, more throttle and right yaw must increase their normalized inputs. Correct RCn_REVERSED first; adjust surface direction later with SERVOx_REVERSED. Recalibrate after endpoint changes. [8]

Modes: initially set all six slots MANUAL. Assign the middle detent to FBWA. Keep the third MANUAL until navigation checks pass, then deliberately assign RTL. Check HUD mode at every detent and transition.

Receiver telemetry is not automatically flight-battery voltage. Keep antenna active ends clear of carbon and power wiring; range-test after final installation.

Sensors and battery readings

Calibrate in final orientation; inspect the actual response

Attitude, heading and position

Attitude: secure FC and set AHRS_ORIENTATION. Calibrate accelerometers in six positions, then level against the aircraft datum. Verify HUD roll/pitch follow the aircraft.

Compass: use the M9N-5883 external compass. Mount level and arrow forward, at least 100 mm from power wires, ESC, motor and ferrous parts. Manufacturer ArduPilot setting: Rotation None. Calibrate and verify known headings; confirm firmware detects the fitted QMC5883L/P. [18]

GPS: use SERIAL3_PROTOCOL=5 and the appropriate GPS type. Check outdoors for plausible position, fix and time. GPS-only hardware does not provide magnetic heading.

Battery monitor - V2 values

Parameter	Initial value
BATT_MONITOR / voltage / current pins	4 / BATT_VOLT_PIN=10 / BATT_CURR_PIN=11
BATT_VOLT_MULT / BATT_AMP_PERVLT	11.0 / 66.7; older-board 31.7 is wrong for V2
BATT_CAPACITY	5200 mAh nominal; validate usable capacity separately

Calibrate against a meter:

New voltage multiplier = old multiplier x meter voltage / reported voltage.

New current scale = old scale x reference current / reported current.

Use a steady meaningful current, not near zero. Check idle offset and consumed capacity. [1,4]

Set low/critical thresholds, delays, reserves and actions from pack specifications and measured loaded behaviour. Record them; the C-rating does not prove endurance.

Servos and V-tail direction

Left/right viewed from behind the aircraft, facing forward

Centre servos electrically before fitting horns. Align linkages mechanically; use SERVOx_TRIM for small corrections. Set conservative endpoints and check every hinge through full travel.

Test	Required response
MANUAL: roll right	Right aileron trailing edge up, left down; reverse for roll left.
MANUAL: pitch up	Both tail trailing edges move in upward pitch-producing sense; reverse for pitch down.
MANUAL: yaw right	Both tail trailing edges move toward aircraft right; reverse for yaw left.
FBWA, centred sticks: tilt aircraft right	Left aileron up, right down: correction opposes tilt.
FBWA: raise nose	Tail command produces nose-down correction; lowering nose produces nose-up correction.

V-tail correction: wrong in both pitch and yaw â†’ reverse output. Wrong in only one axis â†’ exchange functions 79/80 and repeat both tests. Physical motion governs the final mapping. [9]

Roll-to-rudder testing requires nonzero KFF_RDDRMIX; restore its intended value afterward. A stationary yaw test does not prove heading hold. [9]

Combined commands and supply loading

Test all four full pitch/yaw combinations. Check binding, buckling, buzzing and saturation. Adjust mechanics, endpoints and MIXING_GAIN; repeat MANUAL and FBWA tests. [9]

Load-test all servos together without prolonged stall. Capture voltage dips, current peaks, temperature and resets. A multimeter may miss brief dips. Limits: Vx 5 A continuous / 6 A peak; electronics 5 V rail 2 A. Verify margin. [1]

Airspeed and propulsion

Sensor response is not calibration; motor spin is not thrust

Airspeed branch

Use ASPD-4525 on I2C2: ARSPD_TYPE=1, ARSPD_BUS=1. Reboot and confirm airspeed health and compass detection. [1,6]

Put pitot in clean airflow, clear of prop wash. Match dynamic/static ports; avoid leaks and kinks. Let temperature settle. Shield from wind without sealing ports; run Preflight Calibration for zero. [10]

Apply gentle airflow: reading should rise and return near zero. Avoid pressure, liquid or blowing into tubing. Start ARSPD_USE=0; some Plane versions still use a healthy sensor for speed scaling. [10,11]

If readings are suspect, set ARSPD_TYPE=0 and resolve them. Ratio calibration needs a safely flying aircraft and log review. Bench airflow does not establish true airspeed or stall speed.

ESC and motor

Motor mount: four M3 holes on 19 mm PCD; adjacent spacing 13.435 mm. Check actual motor fit and safe screw engagement. The 9-inch /4S /40 A setup is the selected design assumption. Record current, thrust and temperature before flight. [15]

Confirm ESC PWM range and calibration method. FC/RX must stay powered while ESC power switches separately. Use a documented isolated setup or servo tester; do not backfeed power rails. [12]

If the ESC manual specifies high/low calibration: remove prop, switch ESC off, command maximum, power ESC, wait for its indication, then command minimum. Verify acceptance. Do not use multicopter calibration for Plane. [12]

Check stopped motor when disarmed and at minimum. Restrain the airframe for a brief low-speed rotation check. Power off before swapping two phase wires to reverse. Confirm pusher prop orientation produces forward thrust.

Failsafes: prove the trigger

Prop removed; disconnect motor power for configuration tests

With prop removed and telemetry recording, switch the transmitter off. Observe RC inputs, failsafe message, mode and recovery when the link returns.

Radio loss

Set `THR_FAILSAFE=1`. Check whether RX stops frames, flags loss or holds/presets channels. Held normal channels may hide link loss. Configure a detectable condition and repeat. [13]

For low-throttle detection, put `THR_FS_VALUE` between measured loss and valid-low throttle. Example only: loss 900, valid low 1000, threshold 950 us. Require clear separation.

Candidate policy: short loss `CIRCLE`, `FS_SHORT_ACTN=1`; sustained loss `RTL`, `FS_LONG_ACTN=1`. Verify firmware labels and both timeouts. Set valid home, return height, terrain clearance and power reserve before flight. [13]

Battery and navigation failures

Record low/critical voltage, capacity, delay and actions. Test with a suitable variable supply or temporary thresholds, then restore and read back flight values. Never deeply discharge the LiPo for a test.

A ground `RTL` mode change does not prove a safe return. Test navigation-failure policy in simulation, then controlled evaluation with an experienced pilot. Do not unplug live sensors near a running prop.

Record trigger, delay, action and recovery. Repeat from every intended flight mode. If a failsafe cannot be observed or explained, keep the flight gate closed.

Installation and release record

Complete on the actual aircraft

Restrain the adjustable battery. Separate power wiring from GPS, compass and receiver. Protect wires at edges, strain-relieve plugs and label wing/tail connectors. Keep USB, SD, rail bridge and battery access open.

Provisional setup balance station

Provisional balance mark: 522 mm aft of nose, or 22.2 mm aft of front joiner. This is geometric quarter-MAC, not an approved flight CG range. Check complete mass, control authority and final CG with an experienced fixed-wing pilot.

xCG = sum(mass x position) / total mass

Battery adjustment: battery shift = total mass / battery mass x desired CG shift.

Use one nose datum, aft positive. Include wiring, glue and finish. Mark repeatable pack positions; verify assembled balance and lateral balance.

Record	Measured / approved value
Firmware / saved parameters	_____
5 V minimum / Vx minimum / servo peak current	_____
Voltage calibration / current reference	_____
RC loss trigger / short / long / recovery	_____
Battery low / critical / actions	_____
Total mass / battery mass / actual CG from nose	_____
Desired CG shift / required battery movement	_____
Control throws / propulsion test / flight CG review	_____

Before flight: verify connectors, loaded supplies, control directions, range/failsafes, sensors, structure, CG and propulsion. Measure current, thrust, voltage and temperature on a proper restrained test fixture.

SiK telemetry / UART1

433 MHz, stated 100 mW / air radio to FC; USB radio to laptop

The retailer mixes 100 mW and 500 mW specifications. Use the actual radio label and firmware limits. A power setting does not identify the hardware. Fit the correct antennas before applying power.

Air radio function	F405-WING V2 connection
Supply / ground	Regulated 5V / G. Never VBAT or the servo Vx rail.
Radio TX	FC RX1
Radio RX	FC TX1
CTS / RTS, if present	Leave disconnected for the initial no-flow-control wiring; disable flow control if that parameter is exposed.
Antenna	Correct 433 MHz antenna; support it independently so the connector carries no bending load.

Connect and configure

1. With power off, map the supplied cable by signal labels and continuity. A Pixhawk plug is not a Matek pinout. Do not use cable colour as proof.
2. Set SERIAL1_PROTOCOL=2 for MAVLink2 and SERIAL1_BAUD=57 for 57600 baud. Write, reboot and read back. Match the radio serial speed to 57600.
3. Plug the ground radio into the laptop. In Mission Planner SiK Radio settings, read both radios. Match network ID, air rate and radio link settings; save both ends. Keep a supported legal frequency/power configuration.
4. Select the ground radio COM port at 57600 and connect. Tilt the aircraft and verify live attitude, battery readings, GPS and parameter retrieval. A link LED alone is not the full test.
5. Check link statistics and the separate FlySky control link with the installed antennas and all electronics powered. Test telemetry loss separately from RC loss; do not assume they trigger the same failsafe.

Power budget: SiK shares the FC 5 V electronics rail with RX, GPS and airspeed. Include measured radio transmit-current peaks in the 2 A rail test. Keep servo power on Vx/Vx2. The advertised 10 km range is not a demonstrated aircraft range.

Manuals and reference links

Hardware and setup references / checked September 2026

[1] Matek V2 specifications, layout and mapping

<https://www.mateksys.com/?portfolio=f405-wing-v2>

[2] ArduPilot F405-Wing (older details need V2 cross-check)

<https://ardupilot.org/plane/docs/common-matekf405-wing.html>

[3] ArduPilot Choosing Servo Functions

<https://ardupilot.org/plane/docs/servo-functions.html>

[4] ArduPilot board hardware definition (master)

https://github.com/ArduPilot/ardupilot/blob/master/libraries/AP_HAL_ChibiOS/hwdef/MatekF405-Wing/hwdef.dat

[5] FlySky FS-iA10B specification

<https://www.flysky-cn.com/ia10b-canshu>

[6] Matek ASPD-4525

<https://www.mateksys.com/?portfolio=aspd-4525>

[7] Matek ASPD-DLVR

<https://www.mateksys.com/?portfolio=aspd-dlvr>

[8] ArduPilot RC input mapping and reversal

<https://ardupilot.org/plane/docs/rc-throw-trim.html>

[9] ArduPilot V-tail / inverted V-tail setup

<https://ardupilot.org/plane/docs/guide-vtail-plane.html>

[10] ArduPilot Airspeed Calibration

<https://ardupilot.org/plane/docs/calibrating-an-airspeed-sensor.html>

[11] ArduPilot Airspeed Parameters Setup

<https://ardupilot.org/plane/docs/airspeed-parameters-setup.html>

[12] ArduPilot ESC Calibration

<https://ardupilot.org/plane/docs/common-esc-calibration.html>

[13] ArduPilot Plane Failsafe Function

<https://ardupilot.org/plane/docs/apms-failsafe-function.html>

[14] ArduPilot Ground Steering

<https://ardupilot.org/plane/docs/tuning-ground-steering-for-a-plane.html>

[15] Emax ECO II 2807 official drawing in product description

<https://emaxmodel.com/products/emax-eco-ii-series-2807-1300kv-1700kv-1500kv-brushless-motor-for-rc-drone-fpv-racing>

[18] Matek M9N-5883 GNSS and compass

<https://www.mateksys.com/?portfolio=m9n-5883>

[19] TowerPro MG90S specification

<https://towerpro.com.tw/product/mg90s-3/>

[20] ArduPilot SiK telemetry radio

<https://ardupilot.org/plane/docs/common-sik-telemetry-radio.html>

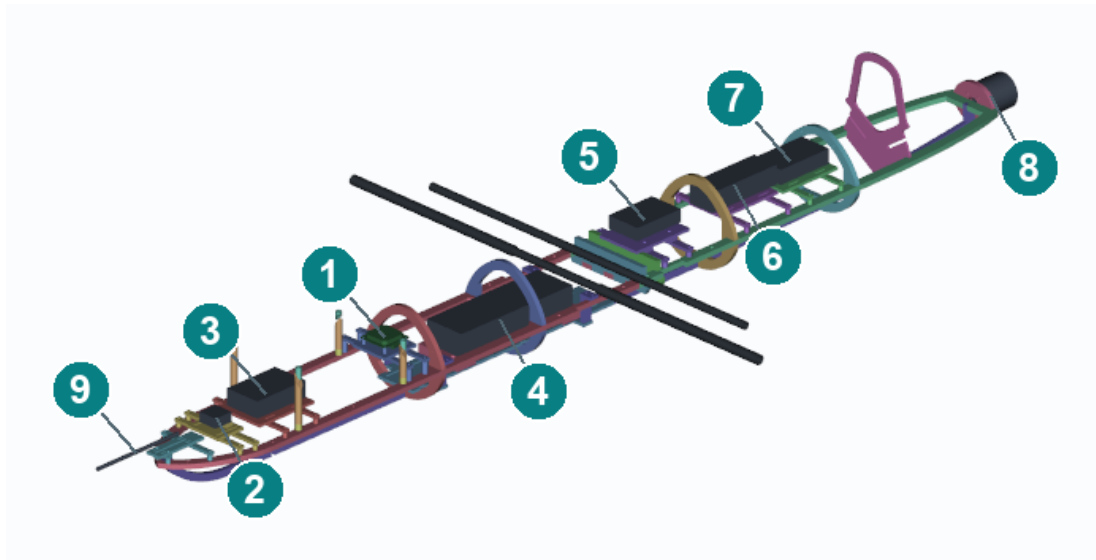
[21] ArduPilot automatic takeoff and hand launch

<https://ardupilot.org/plane/docs/automatic-takeoff.html>

Save the parameter file and connector diagram for the actual aircraft. Complete the loaded-power, sensor, radio and propulsion tests in this guide.

Installed equipment map

Current mounts and access / dimensions are not paper cutting templates



Mark	Placement and attachment
1 / 2 / 9	GPS inside the nose on I09 at Y=-153, antenna up and arrow forward; ASPD-4525 on I13; pitot on I14, aimed forward with pressure ports clear of the nose.
3 / receiver	FS-iA10B on I10 at Y=-300, with independent support for both active antenna tips.
4 / battery	GenX pack on the low central I01 tray. Centre travel Y=-60..+20, or 379.65..459.65 mm aft of the nose. Two straps plus corner stops; battery main and balance leads exit aft.
5 / FC	F405-WING V2 on I04 at Y170, with supplied grommets and 6 mm stand-offs. Arrow/orientation must match setup.
6 / 7	ESC at Y280 on I12; SiK air radio at Y352 on I11. Service through the removable rear foam cover.
8 / motor	Four M3 screws through A26 on the 19 mm bolt circle. Measure engagement and secure the phase leads clear of the bell.

The Manufacturing Team chapter gives fasteners, cover removal and assembly order. The FC is inside the centre fairing; the GPS and airspeed sensor are inside the nose; only the pitot probe projects outside.

Harness routes and balance

Make the installed harness before closing the body

Run	Construction and check
Battery -> FC -> ESC	Use the FC battery/distribution pads, correct polarity and insulated connectors. Start with 12 AWG flexible silicone power wire where extensions are required, matching connector/ESC ratings. Keep DC runs short. Battery leads exit aft; near the GPS bay keep every high-current run at Y=-37 or farther aft, at least 100 mm from the GPS module.
ESC -> motor	Route three phase leads aft along one side of A43. Support extensions and keep them clear of the motor bell, hinges and telemetry antenna. Do not route them through the battery bay.
Servos / wings	Use quality 22-24 AWG servo extensions as required. Label left/right disconnects; support connectors so wing removal cannot pull FC pins. Check loaded voltage at the farthest servo.
GPS / airspeed	Route on the opposite side from high-current wiring. TX/RX cross to their UART partners; SDA/SCL connect in parallel to I2C2. Include ground; insulate and strain-relieve every branch.
Pitot tubing	Keep both tubes distinguishable, unkinked and serviceable. Check the marked total/static ports. Do not blow hard into the sensor or seal a pressure port for calibration.
Antennas	Receiver active tips at right angles, clear of carbon and power leads. Support the 433 MHz antenna separately. Keep GPS/compass clear of ferrous fittings and high-current wiring.

Dry-route wires, add a modest service loop, then measure and cut. Record actual lengths. Extended I2C wiring must pass repeated cold-start detection and a complete motor-on interference check; do not treat a powered LED as a working sensor.

Set balance from measured masses

Replace estimates in Mass_Balance_Worksheet.csv with weighed values. The 522.3 mm aft-nose reference is geometric quarter-MAC, not a flight-approved limit. The sensitivity estimate is about 2.6-3.1 kg, depending on print mass and stated allowances; it is not the aircraft weight.

Balance the complete aircraft, including propeller, connectors and finish. Move the 512 g battery within its 80 mm range, reset both end stops and straps, and recheck lateral balance. If the required location is outside the tray, resolve the weight distribution before flight; do not add unreviewed ballast.

Hand launch and grass landing

Launch method remains unqualified: read Weight and flight assessment

This is a pusher. Prove that motor start occurs only after the aircraft has cleared the launching hand. ArduPilot warns that default takeoff settings can start the motor immediately. [21]

Takeoff control	Review with the installed firmware
TKOFF_THR_MINACC	Launch acceleration trigger. The ArduPilot example is about 15 m/s ² . Check normal handling cannot trigger it and an actual launch motion can.
TKOFF_THR_DELAY	Units are tenths of a second. ArduPilot specifies at least 2, meaning 0.2 s, for pusher hand launches; more may be needed. This is not a guarantee of hand clearance.
TKOFF_THR_MINSPD	Additional GPS-speed gate. The documented starting example is 4 m/s. Check sensor delay and reliability; a gate that never opens is also a failed launch.
TKOFF_THR_IDLE	Keep zero for this stopped-prop launch plan, if the parameter is present. Verify actual motor output at arming and before launch.
Pitch, throttle and modes	Choose TAKEOFF/AUTO behavior, climb limits, abort/recovery and pilot takeover for the measured aircraft. Do not copy generic climb gains or a CG limit as an approved setup.

1. With the motor electrically disconnected and propeller removed, test mode selection, arming, throttle suppression, launch triggering, delay and cancellation. Log the response; repeat after firmware or parameter changes.
2. Before flight, an experienced pilot reviews mass, balance, control authority, thrust, range, failsafes and the launch grip. Use the reinforced fuselage near the measured CG; do not grip a control, foam-only edge, sensor or antenna.
3. Practise the agreed handoff and abort sequence. Use a suitable open area and launch into the wind with wings level. A heavy aircraft must not be treated as a small foam glider merely because it fits in one hand.
4. For landing, use short, even grass and the reviewed power-off approach. Do not rely on a stopped three-blade propeller to eliminate a strike. Check the gear, firewall, propeller and sensor mount after every hard arrival.

Flight-time planning

Estimate from battery capacity / replace assumed current with flight logs

Planning estimate: about 12-20 minutes. This assumes roughly 12-20 A average total current and landing with 20% of the nominal battery capacity unused. Cruise current has not been measured.

Average battery current	Time using 4.16 Ah
8 A	31 minutes
12 A	21 minutes
16 A	16 minutes
20 A	12.5 minutes
25 A	10 minutes
40 A	6 minutes

Calculation

Usable capacity for planning = $5.2 \text{ Ah} \times 0.80 = 4.16 \text{ Ah}$.

Time in minutes = $60 \times 4.16 / \text{average battery current in amperes}$.

Average current must include launch, climb, cruise, servos and avionics. The 40 A ESC rating is not an estimate of cruise current. Static partial-throttle current is not a substitute for a flight log.

Set the first-flight limit

Start with a short evaluation flight and conservative landing reserve. Compare the calibrated current/capacity log with the charger return, voltage under load and the pack condition. Increase duration only after repeated results support it.

The 12-20 minute range is a planning assumption, not an aerodynamic simulation or a demonstrated endurance result. Actual mass, propeller efficiency, speed, wind, manoeuvring and usable battery capacity can move it outside that range.